

BIBLIOTHECA ALEXANDRINA YOUTH CREATORS AWARD 2026 · TECHNOLOGY CREATIVITY TRACK

GeoAI Deforestation Risk Intelligence Platform

An AI-Powered Spatial Decision-Support Platform for Deforestation Risk Assessment in Rondônia, Brazil.

Author & Researcher: Mamdouh Alwakil | **Platform:** amazon-deforestation-risk-3d.vercel.app | **Repository:** github.com/mamdouhahmed5455-lang/Amoazon2

CORE PURPOSE "An AI-assisted spatial decision-support prototype for interpreting and prioritizing relative deforestation risk."

STUDY GEOGRAPHY

**Rondônia,
Brazil**

Legal Amazon Basin

SPATIAL COVERAGE

~150,000 Cells

Precomputed Grid

MODEL PIPELINE

**Offline
XGBoost**

Gradient Boosted Trees

TRACK 1 BENCHMARK

ROC-AUC 0.82

F1-Score: 0.81 (Locked)

TRACK 2 RESEARCH

**ROC-AUC
0.7474**

5.0× Top-Decile Lift

MULTIDISCIPLINARY JURY EVALUATION MATRIX (THE FIVE CORE QUESTIONS)

EXECUTIVE OVERVIEW

INQUIRY

VERIFIED TECHNICAL REALITY & EVALUATION FINDINGS

WHAT is evaluated?

An end-to-end GeoAI decision-support prototype that synthesizes multi-source geospatial and demographic indicators into interpretable, relative spatial risk rankings across the state of Rondônia.

WHERE does it operate?

State of Rondônia, Brazilian Amazon Basin — covering approximately 150,000 precomputed spatial cells, contextualized by 25 years of official INPE PRODES deforestation history (2001–2025) across all 9 Legal Amazon states.

WHAT technology?

Offline-trained XGBoost classification pipeline, spatial feature engineering (STRtree / cKDTree / Copernicus DEM), and a static client-side WebGL (Deck.gl / Mapbox GL JS) visualization engine deployed on Vercel Edge CDN with zero backend database requirement.

WHAT is demonstrated?

Two Distinct Evidence Tracks: Documented Production Benchmark (Track 1: ROC-AUC 0.82, Precision 0.79, Recall 0.84, F1 0.81) + Independent Research Evaluation (Track 2: ROC-AUC 0.7474, 95% CI: [0.5837, 0.9036], Brier 0.00693, 10 positive events, 5.0× Retrospective Top-Decile Enrichment).

WHY does it matter?

The project explores how environmental observations can be organized into a structured workflow for spatial risk screening, prioritization, interpretation, and verification planning under severe logistical and physical resource constraints.

Evidence Classification & Scientific Verdict: Categorized as **MODERATE EVIDENCE**. Empirical out-of-time testing establishes that the model identifies subsequent PRODES clear-cut locations with moderate ranking skill (ROC-AUC 0.7474, 5.0× top-decile lift). All outputs represent **relative spatial risk rankings**, not calibrated probabilities. The prototype serves as an operational screening filter, not a replacement for satellite monitoring systems.

Section 2 • Dual-Track Evidence Architecture & Benchmark Separation

RIGOR PRINCIPLE

"Two evidence tracks, clearly separated: documented production benchmark vs. independent research evaluation."

A foundational principle of this evaluation is the **strict scientific separation** between two distinct evidence tracks. Collapsing these tracks into a single claim or presenting research experiments as exact reproductions of legacy benchmarks is scientifically invalid. Both tracks are documented with complete transparency:

TRACK 1 — DOCUMENTED PRODUCTION BENCHMARK

CANONICAL REFERENCE

The original engineering benchmark documented with the production prototype. Evaluated on a historical partition reflecting the initial project release.

- Model:** Offline-trained XGBoost classification model.
- Metrics:** ROC-AUC 0.82 | Precision 0.79 | Recall 0.84 | F1 0.81.
- Status:** Locked upstream reference asset.
- Scientific Boundary:** Original training code, split seeds, and raw training rasters were unrecorded upstream. Exact reproduction is not claimed.

TRACK 2 — INDEPENDENT RESEARCH MODEL V2

EMPIRICAL VERIFICATION

An independent, fully reproducible empirical research experiment designed to test whether open geospatial features can rank subsequent deforestation risk out-of-time.

- Model:** XGBoost v2.0.0 pipeline trained on neutral lattice.
- Metrics:** ROC-AUC 0.7474 (95% CI: [0.5837, 0.9036]) | PR-AUC 0.0055.
- Evaluation:** Held-out PRODES 2024 target year (10 events).
- Status:** 100% reproducible via repository scripts and 52 tests.

SIDE-BY-SIDE METHODOLOGICAL & EMPIRICAL COMPARISON MATRIX			MULTI-PERSPECTIVE AUDIT
METHODOLOGICAL DIMENSION	TRACK 1: DOCUMENTED PRODUCTION BENCHMARK	TRACK 2: INDEPENDENT RESEARCH MODEL V2	
Spatial Population	~150,000 Web Display Grid Cells (Rondônia)	7,045 Neutral Spatial Units (Multi-year panel: 21,135)	
Feature Inputs	12 documented legacy rasters (unrecorded code)	5 engineered variables across 4 groups (OSM, PRODES, IBGE, DEM)	
Target Ground Truth	Documented historical clear-cut loss	Official INPE PRODES Clear-Cut Polygons (≥ 6.25 ha)	
Temporal Protocol	Static snapshot partition (circa 2020)	Strict out-of-time rolling annual isolation (2022/2023/2024)	
Discrimination (ROC-AUC)	0.8200 (Documented reference)	0.7474 (95% Bootstrap CI: [0.5837, 0.9036])	
Precision / Recall / F1	Prec: 0.79 Rec: 0.84 F1: 0.81	Top-10% Prec: 0.0071 Top-10% Rec: 0.50 (5/10 events)	
Precision-Recall AUC	Unrecorded upstream	0.005534 (Baseline prevalence: 0.001419)	
Calibration Metric	Unrecorded upstream	Brier Score: 0.00693 (Relative ranking; not calibration)	
Top-Decile Enrichment	Unrecorded upstream	5.0× retrospective lift (Captured 50% of test events)	
Scientific Reproducibility	Historical reference (Locked upstream)	100% deterministic code & data (scripts/model_v2/run.py)	

Explicit Scientific Statement: Track 2 (Model V2) is an independent empirical verification research experiment. It is not presented as a reproduction of the Track 1 benchmark and does not replace it. Both tracks are maintained to provide the competition jury with complete transparency regarding what is documented historically versus what has been independently validated.

Section 3 - Authoritative Data Provenance & Experimental Protocol

Model V2 was constructed using exclusively primary, authoritative open datasets acquired directly from Brazilian federal agencies and international scientific repositories. Zero synthetic data was used.

AUTHORITATIVE PRIMARY DATA SOURCES REGISTRY

SOURCE AUDIT

DATA SOURCE	ASSET DESCRIPTION	COORDINATE SYSTEM / FORMAT	ROLE IN EMPIRICAL PIPELINE
INPE TerraBrasilis PRODES	Annual official clear-cut deforestation polygons (2021–2024)	SIRGAS 2000 / EPSG:4674 GeoJSON / Shapefile	Ground-truth binary labels ($y \in \{0, 1\}$) for clear-cuts ≥ 6.25 ha and historical loss distance features.
OpenStreetMap (OSM) Overpass	Highway network corridors (BR-364, BR-421, BR-425, BR-429)	EPSG:4326 GeoJSON line vectors	Euclidean road proximity calculations via STRtree spatial indexing and cKDTree spatial joins.
IBGE 2022 Decennial Census	Municipal populations & municipal seat centroids	Tabular API / EPSG:4326 point coordinates	Demographic gravity potential model representing localized agricultural and settlement pressure.
Copernicus DEM GLO-90	90-meter physical topographic elevation model	Raster GLO-90 / API elevation queries	Terrain accessibility filtering; steep terrain represents a physical barrier to heavy machinery.
INPE Sovereignty & Mask Layers	Rondônia boundary & 3,265 natural non-forest polygons	EPSG:4674 multi-polygons	Sovereign geographic envelope filtering and exclusion of water bodies, savanna (cerrado), and rock outcrops.

MODEL V2 FEATURE ENGINEERING SPECIFICATION (5 VARIABLES ACROSS 4 GROUPS)

FEATURE ENGINE

VARIABLE NAME	PHYSICAL GROUP	MATHEMATICAL DEFINITION	PHYSICAL DOMAIN RATIONALE
dist_road_km	Infrastructure	$\min_k \ x_i - r_k\ _2$ (km to highway)	Paved and unpaved corridors provide access for timber extraction and cattle ranching.
dist_hist_loss_km	Historical Loss	$\min_j \ x_i - h_j\ _2$ (km to loss $\leq T-1$)	Deforestation frontiers expand outward contiguously from existing clearing edges.
prior_loss_dens	Historical Loss	$\sum_j \mathbb{I}(\ x_i - h_j\ \leq 10 \text{ km})$	Aggregated historical loss density measures active frontier momentum.
pop_pressure	Demographic	$\sum_m \text{Pop}_m / (d_{im} + 1)^2$	Urban and municipal centers drive local agricultural commodity demand.
elevation_m	Topography	DEM altitude value at centroid x_i	Lowland flat terrain is preferentially cleared for mechanized agriculture over steep uplands.

NEUTRAL SPATIAL LATTICE DESIGN

Evaluated on a systematic **0.05° (~5.5 km) geographic grid** across Rondônia. Every cell centroid was strictly validated: must lie within state sovereignty AND outside all 3,265 non-forest mask polygons. This yields exactly **7,045 eligible forest units** evaluated annually (**21,135 multi-year panel observations**). Zero score inheritance from production files was permitted.

TEMPORAL SPLIT PROTOCOL & CLASS SPARSITY

Annual Monitoring Partitions:

- *Training (2022)*: 7,045 cells | 41 positives | 0.5820% prevalence (1 in 172)
 - *Validation (2023)*: 7,045 cells | 23 positives | 0.3265% prevalence (1 in 306)
 - *Held-Out Test (2024)*: 7,045 cells | 10 positives | 0.1419% prevalence (1 in 705)
- Reflects authentic real-world rare-event physical class imbalance.

Section 4 • Model Explainability: TreeSHAP vs. Global Analytical Evidence

To ensure responsible AI deployment, the platform maintains two distinct interpretability frameworks. They answer different scientific questions and operate on different data populations. Neither framework represents linear regression coefficients or causal effect sizes.

BLOCK A • MODEL V2 TREESHAP ATTRIBUTION

EMPIRICAL SHAP

Computed via exact TreeSHAP across all **7,045 held-out test units** (PRODES 2024). Measures the marginal algorithmic contribution of each feature to Model V2 log-odds predictions.

FEATURE	PHYSICAL GROUP	MEAN SHAP	SHARE %
dist_hist_loss_km	Historical Loss	1.4130	41.28%
dist_road_km	Road Proximity	0.5969	17.44%
elevation_m	Topography	0.5604	16.37%
pop_pressure	Demographic	0.4266	12.46%
prior_loss_dens	Historical Loss	0.4260	12.45%

• **Combined Historical Loss Impact: 53.73%** (distance + density).

• In this empirical 2024 test partition, proximity to prior clear-cuts accounted for the largest share of decision-tree attributions.

BLOCK B • DOCUMENTED GLOBAL ANALYTICAL IMPORTANCE

TRACK 1 REFERENCE

Documented global feature-importance evidence from the original platform documentation. Reflects macro-structural regional driver attribution.

FEATURE DRIVER	ANALYTICAL IMPORTANCE	VISUAL DISTRIBUTION
Road Proximity	41.0%	
Forest Loss (History)	23.0%	
Population Pressure	21.0%	
Elevation Constraints	15.0%	

• **Road Infrastructure Dominance (41%):** Highlights highway corridors (BR-364) as the historical conduits of Amazonian frontier expansion.

• **Sum of Weights: Exactly 100%** across the four canonical drivers.

COMPARATIVE ANALYSIS: WHY THE TWO EXPLAINABILITY VIEWS DIFFER			SCIENTIFIC INTERPRETATION
DIMENSION	BLOCK A: EMPIRICAL MODEL V2 TREESHAP	BLOCK B: DOCUMENTED GLOBAL ANALYTICAL IMPORTANCE	
Analytical Scope	Marginal algorithmic split contribution on 2024 test data	Macro-structural regional frontier driver attribution	
Primary Driver	Historical Loss Proximity (41.28% / 53.73% combined)	Road Corridor Proximity (41.00%)	
Why the Shift Occurred	In the 2024 test set, clear-cuts occurred adjacent to previous clearings; historical loss was the strongest immediate spatial separator.	Over multi-decade timescales, road networks provide the macro accessibility that enabled clearings to occur in the first place.	
Mathematical Status	Shapley values based on game theory (shap.TreeExplainer)	Documented analytical weighting benchmark (canonical reference)	

Critical Methodological Notice: TreeSHAP attribution is specific to Track 2 (Research Model V2) and is not claimed as the explainability method for the Track 1 production benchmark. Neither set of values should be interpreted as causal effect sizes or policy intervention levers.

GeoAI Deforestation Risk Intelligence Platform • Technical Evidence Dossier

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Section 5 • Empirical Validation, Retrospective Top-K Enrichment & Ablation Findings

ROC-AUC (TEST 2024)
0.7474
95% CI: [0.5837, 0.9036]

PR-AUC
0.005534
Baseline: 0.001419 (3.9×)

Brier Score
0.00693
Mean Squared Error

TOP-10% LIFT
5.0×
5/10 Test Events Hit

OBSERVED POSITIVES
10 Cells
N = 7,045 (0.1419%)

Four-Stage Incremental Ablation Study & Single-Feature Baselines				Feature Ablation
Model Configuration	Features	Variables Included	ROC-AUC	PR-AUC
Model A (Road Only)	1 var	dist_road_km	0.4495	0.001833
Model B (Road + Loss)	3 vars	dist_road_km, dist_hist_loss_km, prior_loss_dens	0.7349	0.003753
Model C (Road + Loss + Pop)	4 vars	Model B + pop_pressure	0.7500	0.008011
Model D (All 4 Groups — Full V2)	5 vars	Model C + elevation_m	0.7474	0.005534
Historical-Loss-Only Benchmark	1 var	dist_hist_loss_km (Inverse distance)	0.7797	0.004900

Key Scientific Finding: Showing that proximity to historical loss was the strongest single-feature benchmark in this evaluation sample and outperformed the full multi-factor V2 model. Historical clear-cut edges exhibit strong localized spatial inertia. The value of multi-feature integration (Model V2) lies in stabilizing spatial risk rankings across diverse terrain and sub-regions where simple proximity heuristics fail.

Retrospective Top-K Prioritization					Spatial Subregion Diagnostics (Small-Sample)			
Inspection Tier	Cells	Hits / Total	Recall	Lift	Diagnostic Partition	Samples	Positives	ROC-AUC
Top 1.0%	70	0 / 10	0.0%	0.0×	Southern Region (Interior)	3,268	4 events	0.9092
Top 5.0%	352	2 / 10	20.0%	4.0×	Northern Region (Frontier)	3,777	6 events	0.6109
Top 10.0%	704	5 / 10	50.0%	5.0×	Cross: North '22 → South '24	3,268	4 events	0.8795
Top 20.0%	1,409	6 / 10	60.0%	3.0×	Cross: South '22 → North '24	3,777	6 events	0.5614
*Retrospective ranking evidence across 2024 test events, not a measured field operational efficiency gain.					*Exploratory spatial diagnostics based on small positive samples (4–6 events); interpret with caution.			

Interpretation Boundary: Brier score (0.00693) reflects low mean squared error driven by extreme class rarity (0.14% positive prevalence). It does **not** prove probabilistic calibration. Model V2 outputs are relative spatial risk rankings, not calibrated probabilities.

Section 6 • Reproducibility Audit & The Scientific Halt Notice

INTEGRITY RULE "A scientifically valid halt due to missing evidence is better than a fabricated reproduction."

A hallmark of institutional scientific maturity is the willingness to document what could *not* be reproduced, rather than forcing artificial alignment. A rigorous forensic audit classified all repository modeling assets into three clear tiers:

Component-by-Component Reconstruction Boundary Matrix			Forensic Audit
Pipeline Component	Classification	Specific Technical Status & Scientific Boundary	
A. Production Spatial Grid	Tier 1: Reproducible	data/forest_data_clean.json (150,000 cells), coordinates, precomputed scores, and extrusion values.	
B. Historical Time Series	Tier 1: Reproducible	data/prodes_historical.json (2001–2025 series across all 9 Legal Amazon states).	
C. WebGL Frontend	Tier 1: Reproducible	Fully functional static client application running in modern WebGL browsers with zero backend server.	
D. Model V2 Pipeline	Tier 1: Reproducible	Complete end-to-end Python pipeline (scripts/model_v2/run.py) executing deterministically with fixed seed.	
E. Feature Reconstruction	Tier 2: Assumptions	Conceptual features (Roads, Forest Loss, Population, DEM) reconstructed via authoritative public APIs.	
F. Benchmark Training Code	Tier 3: Blocked	Original training scripts, split seed, and raw input rasters were unrecorded upstream; reproduction halted.	
G. Benchmark Ground-Truth	Tier 3: Blocked	Original binary clear-cut labels ($y \in \{0, 1\}$) were completely absent from the web repository.	

The Scientific Halt Decision: Why We Refused to Fabricate Pseudo-Labels

Protocol Halt

When forensic discovery confirmed that original ground-truth binary labels ($y \in \{0, 1\}$) were absent from the repository, the research team faced a critical choice:

A common superficial shortcut is creating synthetic pseudo-labels by thresholding the model's own predictions (e.g., $y_{\text{fake}} = 1$ if $\text{risk_score} \geq 170$). **We formally rejected this practice as scientific misconduct:**

- Self-Referential Target Leakage:** Training on thresholded outputs trains a model to predict its own prior outputs, bypassing real-world satellite ground truth entirely.
- Deceitful Validation:** The resulting metrics would measure surrogate approximation error, not true deforestation detection performance.
- Proper Scientific Resolution:** The team instead constructed Model V2 as a clean, independent experiment using authentic INPE PRODES clear-cut polygons acquired from TerraBrasilis.

Automated Test Suite Coverage (100% Deterministic Pass Rate)			99 Total Tests
Python Model V2 Suite: 52 Tests Passing (~51.1s) Verifies feature bounds, coordinate envelopes, temporal isolation, zero data leakage, and training determinism.	Node.js Unit Tests: 38 Tests Passing Validates feature importance sums (100%), score normalizations [0, 100], scenario directionality, and priority tiers.	Historical PRODES Suite: 9 Tests Passing Verifies 25-year series integrity (2001–2025), Legal Amazon state sums (6,518 / 5,731 km²), and trend classifications.	

Section 7 · Scientific Boundaries, Operational Context & Integrity Statement

VISION PRINCIPLE "Prototype today, scalable environmental intelligence tomorrow."

EXPLICIT SCIENTIFIC BOUNDARIES: WHAT THE PLATFORM IS AND IS NOT

CLAIM DISCIPLINE

DIMENSION	WHAT THE PLATFORM IS (DEMONSTRATED SCOPE)	WHAT THE PLATFORM IS NOT (EXPLICIT NON-CLAIMS)
Telemetry & Satellites	Multi-source static geospatial data synthesis.	NOT live telemetry from NASA FIRMS or Sentinel-2 streaming.
Machine Learning Execution	Precomputed offline XGBoost model inference.	NOT live runtime training or inference executed in the browser.
Risk Score Semantics	Relative spatial risk rankings across ~150,000 cells.	NOT calibrated frequentist probabilities or event forecasts.
Enforcement Actions	Strategic decision-support for patrol prioritization.	NOT autonomous dispatch or legal enforcement instruments.
Institutional Status	Independent scientific and technology competition entry.	NO formal partnerships claimed with IBAMA, ICMBio, or SEDAM.
3D Map Extrusion	Aesthetic visual height representation ($risk_{norm} \times 4000$).	NOT true physical terrain altitude in the 3D map extrusion.
Economic / Carbon Impact	Theoretical capacity-constrained lift calculations.	NO measured carbon savings , financial ROI, or revenue claimed.

PREREQUISITES FOR FUTURE OPERATIONAL DEPLOYMENT (RESEARCH ROADMAP)

ROADMAP

- **Institutional Governance:** Formal data-sharing protocols and role-based access with sovereign environmental authorities.
 - **Automated Satellite Ingestion:** Operational pipelines for real-time DETER optical/radar alert streaming.
- **Edge Field Interoperability:** Offline export to ruggedized field GIS platforms (QField, Garmin GPX, GeoPackage).
 - **Cross-Biome Validation:** Formal model transferability testing in Cerrado, Pantanal, and Pan-Amazonian biomes.

FORMAL EVIDENCE INTEGRITY & SCIENTIFIC TRANSPARENCY COMMITMENT

This Technical & Scientific Evidence Dossier represents an exhaustive, unembellished record of the GeoAI Deforestation Risk Intelligence Platform as preserved in the public repository. All metrics, sample sizes, confidence intervals, ablation drops, and scientific limitations have been reported with complete fidelity. No negative findings have been concealed, no ungrounded capabilities have been exaggerated, and no speculative operational deployments have been claimed.

Principal Developer & Researcher: Mamdouh Alwakil
Spatial Data Engineering · Machine Learning · WebGL Geovisualization

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Technology Creativity Track